STAT 1110 – Summary Tables: One Variable

Some Common Symbols and their Pronunciation

π	μ	$\widehat{p}$	$\bar{x}$	H_0
"pie" or "pee"	"mew" or "moo"	"p-hat"	"x-bar"	"H-naught"

Guiding Question

Is the question dealing with a categorical variable (proportions) or a quantitative variable (means)?

	Categorical Variable (One Proportion: Chapters 1 & 3)	**Quantitative Variable** (One Mean: Chapters 2 & 3)
Graphical Summary	Bargraph	Histogram, Dotplot, Boxplot
Summary Measures	$\widehat{p}$ = sample proportion n = sample size	$\bar{x}$ = sample proportion s = sample standard deviation n = sample size
Summary Statistic	$\widehat{p}$ = sample proportion	$\bar{x}$ = sample mean
Parameter	π = population/long-run proportion (This is the Greek letter "pi")	μ = population/long-run mean (This is the Greek letter "mu")
Null Hypothesis	$H_0 : \pi = \pi_0$ (i.e. the null value)	$H_0 : \mu = \mu_0$ (i.e. the null value)
Alternative Hypothesis (Pick One)	H_A: $\pi > \pi_0$ (one-sided) H_A: $\pi < \pi_0$ (one-sided) H_A: $\pi \neq \pi_0$ (two-sided)	H_A: $\mu > \mu_0$ (one-sided) H_A: $\mu < \mu_0$ (one-sided) H_A: $\mu \neq \mu_0$ (two-sided)
Null Distribution	Distribution of sample proportions, assuming the null hypothesis is true	Distribution of sample means, assuming the null hypothesis is true
Validity Conditions for a Theory-Based Approach	At least 10 successes **and** at least 10 failures	Symmetric **population distribution** **–OR–** $n > 20$ **and** the sample distribution is not heavily skewed
Standardized Statistic	$z = \frac{\widehat{p} - (\text{hypothesized mean of null distribution})}{SD_{NULL}}$ **–OR–** $z = \frac{\widehat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$	$t = \frac{\bar{x} - (\text{hypothesized mean of null distribution})}{SD_{NULL}}$ **–OR–** $t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$
95% Confidence Interval (Using the 2SD Method)	$\widehat{p} \pm 2 \cdot SD_{NULL}$ **–OR–** $\widehat{p} \pm 2 \cdot \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$	$\bar{x} \pm 2 \cdot SD_{NULL}$ **–OR–** $\bar{x} \pm 2 \cdot \frac{s}{\sqrt{n}}$
SD_{NULL}	Obtained from the One Proportion applet	Obtained from the One Mean applet
SE (Standard Error)	$SE(\widehat{p}) = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$	$SE(\bar{x}) = \frac{s}{\sqrt{n}}$